

Project Initialization and Planning Phase

Date	14 July 2026
Team ID	SWUID2026221635
Project Title	A Comprehensive Measure of Well-Being (HDI Prediction & Analysis)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To revolutionize the human development index assessment process by implementing advanced machine learning techniques, ensuring instantaneous, predictive, and highly accurate country tier classifications.
Scope	The project comprehensively evaluates global human development dimensions—incorporating life expectancy, educational attainment, and GNI per capita metrics—into a robust Machine Learning and web framework for streamlined data simulation.
Problem Statement	
Description	Traditional development analysis tools lack predictive foresight, requiring hours of manual computation on static, unorganized datasets. This introduces analytical inefficiencies and hampers a policymaker's ability to simulate structural interventions in real time.
Impact	Solving these issues will result in accelerated data processing, sharper insight into national development gaps, and a robust enhancement in strategic policy planning.
Proposed Solution	
Approach	Employing supervised Machine Learning classification techniques to ingest, clean, and predict HDI tiers dynamically, creating a responsive predictive architecture.
Key Features	• Implementation of a Machine Learning-based human development classification engine. • Real-time decision simulation using adjustable input sliders. • Interactive data visualization graphs mapping development deficits.

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ResourceRequirements

ResourceType	Description	Specification/Allocation
Hardware		
Computing Resources	CPU requirements to process the data and run the Flask web application locally.	Intel Core i3 Processor or above (minimum 2.0 GHz)
Memory	RAM allocated for smooth data handling in Pandas and executing machine learning models without system lag.	Minimum 4 GB RAM.
Storage	Free disk space required to install Anaconda, store project code repositories, and save the dataset.	Minimum 10 GB free space (SSD preferred).
Software		
Frameworks	Lightweight Python web framework chosen to build the application routes and handle user interface inputs.	Flask Framework.
Libraries	Fundamental Python packages used for data preprocessing, training the predictive model, and plotting analysis graphs.	scikit-learn, pandas, numpy, matplotlib, seaborn.
Development Environment	The Integrated Development Environment (IDE) used to write, execute, and debug the code.	Anaconda Navigator / Jupyter Notebook or VS Code.
Data		
Data	The source global database containing multidimensional features like life expectancy, schooling, and income.	UN Human Development Report Dataset, CSV format.